

Work through the following exercises while the TA circulates. When you have completed the exercises, spend the rest of the laboratory working on the homework assignment.

Warm up.

1. Explain the difference between the general anti-derivative of a function $f(x)$ and the expression $\int f(x)dx$.
2. Compute the general anti-derivative of the following functions.
 - (a) $f(x) = 0$.
 - (b) $f(x) = -2$.
 - (c) $f(x) = 4x$.
 - (d) $f(x) = x^n$ with $n \neq -1$.
 - (e) $f(x) = \frac{1}{x}$.
 - (f) $f(x) = e^x$.
 - (g) $f(x) = \sin(x)$.
 - (h) $f(x) = \cos(x)$.

Exercise 1.

Compute the following integrals.

1. $\int_0^{\pi/3} (2\sin(x) + \cos(x))dx$.
2. $\int_{-3}^{-1} \frac{x+x^2}{x}dx$.

Exercise 2.

A function $f(x)$ satisfies $\int_{-2}^0 f(x) = 2$ and $\int_0^1 f(x) = -1$.

1. Sketch the graph of a function $f(x)$ satisfying the two expressions above.
2. Compute $\int_{-2}^1 f(x)$.
3. Sketch the graph of a function $g(x)$ satisfying $\int_{-2}^1 g(x) = 1$ and $g(x) \neq f(x)$ for $-2 \leq x \leq 1$.
4. Explain the difference between $f(x)$ and $g(x)$.